

Brightlearn Exercise 01 SQL Foundation

01 Selecting and Sorting

01 Retrieve all columns from the employees table.

Table employees

id	first-name	last-name	department	salary	hire-date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-12	Durban
4	David	Mukwena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Malden	IT	72000	2022-04-30	Johannesburg
6						

Syntax for the table

SELECT *

FROM employees;

B

02 Find all unique departments

Syntax

DISTINCT

SELECT department

FROM employees;

Result table/column

department
IT
HR
Finance
Marketing

03 Retrieve first and last names ordered by salary descending.

Syntax

Q3) SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;

Results table - employees

first_name	last_name
Eva	Maidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

Q4 Remove the top 3 highest-paid employees;

Syntax

SELECT id, first_name, last_name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;

Results table - employee

id	first_name	last_name	salary
5	Eva	Maidoo	12000
1	Alice	Green	7000
3	Cathy	Zulu	6500

Filtering with WHERE

SQL

Q5) Find employees in the IT department.

Syntax

```
SELECT id, first_name, last_name, department  
FROM employees  
WHERE department = 'IT';
```

Results table - employees

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Nardao	IT

② Find employees in finance with salary > 60000.

Syntax

```
SELECT id, first_name, last_name, department, salary  
FROM employees  
WHERE department = 'finance' AND salary > 60000;
```

Result Table - employees

id	first_name	last_name	department	salary
3	Cathy	Zulu	finance	65000

⑦ Find employees in HR or Marketing

Syntax

```
SELECT id, first_name, last_name, department  
FROM employees  
WHERE department = 'HR' OR department = 'marketing';
```

Results table

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

(08) Find employees Not in IT

```
SELECT *  
FROM employees  
WHERE NOT department = 'IT'
```

Result table employee

id	first_name	last_name	department	salary	email
2	Brian	Lee	HR	45000	✗
3	Cathy	Zulu	Finance	65000	
4	David	Mokwana	Marketing	50000	

09) Find employees in IT, HR or finance using IN

```

SELECT Id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');

```

Results table - employees

Id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zula	Finance
5	Eva	Maldoo	IT

10 Find employees in IT with salary > 65000 and city Johannesburg

Syntax

```

SELECT Id, first_name, last_name, department, salary, city
FROM employees
WHERE department = 'IT' AND salary > 65000 AND city = 'Johannesburg'

```

Results table - employees

Id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Maldoo	IT	72000	Johannesburg